EECS 182/282A Final Review Session
Key Concepts Students Find Confusing

Welcome to the Review Session

Let's focus on the concepts that tend to trip students up on exams.

1 Confusion Point #1: Why RMS norms and not just spectral norms?

The exam question often looks like: "Explain why RMS→RMS operator norms are useful for neural network parameterization."

The key insight you need:

The RMS→RMS norm naturally incorporates dimension scaling:

$$\|A\|_{\text{RMS}\to\text{RMS}} = \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}}\|A\|_2$$

This makes it **width-aware** automatically. Different layers get different effective learning rates from a *single global hyperparameter*.

Exam tip: If you see "layer-specific learning rates" or "fan-in/fan-out scaling" in a question, think RMS→RMS norms.

2 Confusion Point #2: The rank-1 gradient argument

Students often get confused here: Why does batch size 1 give rank-1 gradients?

For a linear layer $h = Wx$:

$$\nabla_W \ell = \frac{\partial \ell}{\partial h} x^T$$

This is an **outer product** of two vectors → rank 1, always.

Critical follow-up for signSGD:

- For rank-1 matrices: $\|A\|_2 = \|A\|_F$ (spectral = Frobenius)
- For $S = \text{sgn}(G)$ with all entries ± 1: $\|S\|_F = \sqrt{d_{\text{out}} \cdot d_{\text{in}}}$
- Therefore: $\|S\|_2 = \sqrt{d_{\text{out}} \cdot d_{\text{in}}}$
- Plugging into RMS→RMS: $\|\eta S\|_{\text{RMS}\to\text{RMS}} = \eta d_{\text{in}}$
- **Conclusion:** $\eta \propto 1/d_{\text{in}}$ to keep updates stable

Exam trap: Don't confuse this with the general case where $\|A\|_2 \le \|A\|_F$ (the inequality is strict for higher-rank matrices).

3 Confusion Point #3: What is μP actually trying to do?

The conceptual framework:

Think of the Central Limit Theorem analogy:

- $\frac{1}{n}\sum z_i \to 0$ (vanishes)
- $\sum z_i$ (blows up)
- $\frac{1}{\sqrt{n}}\sum z_i \Rightarrow \mathcal{N}(0,1)$ (stable)

μP does the same thing for neural networks:
Choose parameterization so that as width $n \to \infty$:

- Activations stay $\Theta(1)$: $\|h_\ell\|_{\text{RMS}} = \Theta(1)$
- **Updates** stay $\Theta(1)$: $\|\Delta h_\ell\|_{\text{RMS}} = \Theta(1)$

Why this matters: Hyperparameters transfer across widths! Train small, deploy big.
Sufficient condition (memorize this):

$$\boxed{\|W_\ell\|_{\text{RMS}\to\text{RMS}} = \Theta(1), \quad \|\Delta W_\ell\|_{\text{RMS}\to\text{RMS}} = \Theta(1)}$$

4 Confusion Point #4: Why does Shampoo $\to UV^T$?

The derivation students miss:

Starting from $G = U\Sigma V^T$, Shampoo (without accumulation) computes:

$$(GG^T)^{-1/4}G(G^TG)^{-1/4}$$

Substituting:

- $GG^T = U\Sigma^2U^T$
- $G^TG = V\Sigma^2V^T$

So:

$$(U\Sigma^{-1/2}U^T)(U\Sigma V^T)(V\Sigma^{-1/2}V^T) = U\Sigma^{-1/2}\Sigma\Sigma^{-1/2}V^T = UV^T$$

The punchline: All singular values $\to 1$ (condition number = 1, most uniform step possible)
Connection to constrained optimization: This is *exactly* what you get from $\min_{\|\Delta W\|_2 \le \eta}\langle G, \Delta W\rangle$

5 Confusion Point #5: How does Newton-Schulz work?

The polynomial trick:

Odd polynomials of the form $p(X) = a_0X + a_1X(X^TX) + \cdots$ **preserve singular vectors**:

$$\text{If } X = U\Sigma V^T, \text{ then } p(X) = Up(\Sigma)V^T$$

The classic iteration:

$$X_{k+1} = \frac{3}{2}X_k - \frac{1}{2}X_kX_k^TX_k$$

This pushes $\sigma \to 1$ (for $\sigma \in (0,1)$).
Critical preprocessing: Normalize by Frobenius norm first!

$$X_0 = \frac{B}{\|B\|_F}$$

guarantees all $\sigma \le 1$ (since $\|B\|_2 \le \|B\|_F$).

Exam question pattern: "Why is normalization necessary?" $\to$ Answer: to ensure stability ($\sigma \in$ convergence basin).

6 Exam Strategy Checklist

Dimension tracking: Always check if d_{in} and d_{out} appear correctly in scaling arguments

Rank awareness: Batch size 1 $\rightarrow$ rank 1 gradients $\rightarrow$ special properties

SVD intuition: UV^T = "all singular values $\rightarrow$ 1" = semi-orthogonal = condition number 1

μP = stable scaling: Both activations *and updates* stay $\Theta(1)$

Newton-Schulz = polynomial on Σ: Changes singular values, preserves U and V

Final thought: If you see a problem asking "why does this scale with width," immediately think: *What norm is being used? Is it RMS-based? How do the dimensions appear?* That's your entry point.

Good luck on the final!

7 Summary of All Conclusion Equations

7.1 Norms and Their Relationships

Vector RMS norm:

$$\boxed{\|x\|_{\text{RMS}} = \sqrt{\frac{1}{d}\sum_{i=1}^{d} x_i^2} = \frac{1}{\sqrt{d}}\|x\|_2}$$

Induced RMS→RMS norm:

$$\boxed{\|A\|_{\text{RMS}\to\text{RMS}} = \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}}\|A\|_2}$$

Rank-1 property:

$$\boxed{\text{For rank-1 matrices: } \|A\|_2 = \|A\|_F}$$

7.2 Spectral-Constrained Optimization

$$\boxed{\Delta W^* = -\eta \cdot UV^T \quad \text{where } G = U\Sigma V^T}$$

7.3 RMS→RMS Constraint

$$\boxed{\|\Delta W\|_{\text{RMS}\to\text{RMS}} \leq \gamma \quad \Leftrightarrow \quad \|\Delta W\|_2 \leq \gamma\sqrt{\frac{d_{\text{out}}}{d_{\text{in}}}}}$$

7.4 SignSGD Scaling (Rank-1 Case)

$$\boxed{\|S\|_2 = \sqrt{d_{\text{out}} \cdot d_{\text{in}}} \quad \text{where } S = \text{sgn}(G)}$$

$$\boxed{\|\eta S\|_{\text{RMS}\to\text{RMS}} = \eta \cdot d_{\text{in}}}$$

$$\boxed{\eta \leq \frac{\gamma}{d_{\text{in}}} \quad \text{for stability}}$$

7.5 μP Desiderata

$$\boxed{\begin{array}{ll} \text{(D1)} & \|h_\ell\|_{\text{RMS}} = \Theta(1) \\ \text{(D2)} & \|\Delta h_\ell\|_{\text{RMS}} = \Theta(1) \end{array}}$$

$$\boxed{\begin{array}{ll} \text{Sufficient:} & \|W_\ell\|_{\text{RMS}\to\text{RMS}} = \Theta(1) \\ & \|\Delta W_\ell\|_{\text{RMS}\to\text{RMS}} = \Theta(1) \end{array}}$$

7.6 Shampoo Equivalence (No Accumulation)

$$\boxed{(GG^T)^{-1/4}G(G^TG)^{-1/4} = UV^T}$$

Derivation:

$$\boxed{\begin{aligned} &= (U\Sigma^{-1/2}U^T)(U\Sigma V^T)(V\Sigma^{-1/2}V^T) \\ &= U\Sigma^{-1/2} \cdot \Sigma \cdot \Sigma^{-1/2}V^T \\ &= UV^T \end{aligned}}$$

7.7 Newton-Schulz Polynomials

Classic:

$$\boxed{p(\sigma) = \frac{3}{2}\sigma - \frac{1}{2}\sigma^3}$$

Practical (NanoGPT):

$$\boxed{f(\sigma) = 3.4442\sigma - 4.7750\sigma^3 + 2.0315\sigma^5}$$

Matrix form:

$$\boxed{X_{k+1} = \frac{3}{2}X_k - \frac{1}{2}X_k X_k^T X_k}$$

Key property:

$$\boxed{\text{If } X = U\Sigma V^T, \text{ then } p(X) = U \cdot p(\Sigma) \cdot V^T}$$

7.8 Muon Normalization

$$\boxed{X_0 = \frac{B_t}{\|B_t\|_F} \quad \Rightarrow \quad \sigma_i(X_0) \leq 1 \text{ for all } i}$$

Why: $\|B\|_2 \leq \|B\|_F$ always holds

8 Comprehensive Methods Summary Table

Table 1: Complete Summary of Optimization Methods

Method	Norm/Constraint(s)	Key Scaling Relations	Solution Form(s)	Key Assumptions	Main Benefit	Cost
General Recipe	$\|\Delta W\| \leq \eta$	Depends on choice of norm	$\Delta W^* = -\eta \cdot \frac{\nabla W}{\|\nabla W\|_*}$	First-order approximation valid	Flexible framework	Varies
Spectral-constrained	$\|\Delta W\|_2 \leq \eta$	Width-agnostic	$\Delta W^* = -\eta \cdot UV^T$ where $G = U\Sigma V^T$	SVD available or computable	• Isotropy (cond. # = 1) • Removes singular value anisotropy	$O(d^3)$ for SVD
RMS→RMS	$\|\Delta W\|_{\text{RMS}\to\text{RMS}} \leq \gamma$ **Equivalent to:** $\|\Delta W\|_2 \leq \gamma\sqrt{\frac{d_{\text{out}}}{d_{\text{in}}}}$	$\|\Delta W\|_{\text{RMS}\to\text{RMS}} = \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}}\|\Delta W\|_2$ $\eta_{\text{eff}} \propto \sqrt{d_{\text{out}}/d_{\text{in}}}$	Same as spectral-constrained but with layer-specific η_{eff}	Fan-in/out known	• Single global $\gamma \to$ layer-specific rates • Width-aware automatically	Same as chosen method
SignSGD (BS=1)	$\|\Delta W\|_{\text{RMS}\to\text{RMS}} \leq \gamma$ **Requires:** $\eta \leq \frac{\gamma}{d_{\text{in}}}$	For $S = \text{sgn}(G)$: $\|S\|_2 = \sqrt{d_{\text{out}} \cdot d_{\text{in}}}$ $\|\eta S\|_{\text{RMS}\to\text{RMS}} = \eta \cdot d_{\text{in}}$	$W_{t+1} = W_t - \eta \cdot \text{sgn}(g_t)$	• Batch size = 1 • Gradient is rank-1 • $\|G\|_2 = \|G\|_F$ for rank-1	• Simple implementation • $\eta \propto 1/d_{\text{in}}$ scaling • Memory efficient	$O(d^2)$ (element-wise)
μP (Maximal Update Param)	**Desiderata:** (D1) $\|h_\ell\|_{\text{RMS}} = \Theta(1)$ (D2) $\|\Delta h_\ell\|_{\text{RMS}} = \Theta(1)$ **Sufficient:** $\|W_\ell\|_{\text{RMS}\to\text{RMS}} = \Theta(1)$ $\|\Delta W_\ell\|_{\text{RMS}\to\text{RMS}} = \Theta(1)$	Uses recursion: $\|\Delta h_\ell\|_{\text{RMS}} \lesssim \|\Delta W_\ell\|_{\text{RMS}\to\text{RMS}} \cdot \|h_{\ell-1}\|_{\text{RMS}} + \|W_\ell\|_{\text{RMS}\to\text{RMS}} \cdot \|\Delta h_{\ell-1}\|_{\text{RMS}}$	Not a specific optimizer; a **parameterization philosophy** Can combine with any optimizer	• RMS norms controlled • Activations stay $O(1)$	• Hyperparameters transfer across widths • Train small, deploy large • Feature learning regime	N/A (parameterization)
Shampoo (Full)	Implicit via preconditioning	**Accumulators:** $L_t = L_{t-1} + G_t G_t^T$ $R_t = R_{t-1} + G_t^T G_t$	$W_{t+1} = W_t - \eta \cdot L_t^{-1/4} G_t R_t^{-1/4}$	• Matrices invertible • Sufficient statistics accumulate	• Adaptive to gradient geometry • Reduces anisotropy	$O(d^3)$ per step (matrix powers)

Continued on next page

Table 1 – continued from previous page

Method	Norm/Constraint(s)	Key Scaling Relations	Solution Form(s)	Key Assumptions	Main Benefit	Cost
Shampoo (No Accum.)	Same as spectral-constrained	With $L = GG^T$, $R = G^TG$: $(GG^T)^{-1/4} = U\Sigma^{-1/2}U^T$ $(G^TG)^{-1/4} = V\Sigma^{-1/2}V^T$	$\Delta W = -\eta \cdot (GG^T)^{-1/4}G(G^TG)^{-1/4}$ **Simplifies to:** $\Delta W = -\eta \cdot UV^T$	$G = U\Sigma V^T$ (SVD)	• Shows equivalence to spectral-constrained • Condition number = 1 • All singular values $\to 1$	$O(d^3)$ (SVD + matrix ops)
Muon (Full)	$\\|X_K\\|_2 \approx 1$ after K iterations **Initial:** $\sigma_i(X_0) \in (0,1)$ for stability	**Normalization:** $X_0 = \frac{B_t}{\\|B_t\\|_F}$ Since $\\|B\\|_2 \leq \\|B\\|_F$: $\sigma_i(X_0) \leq 1$	**Algorithm:** $B_t = \mu B_{t-1} + G_t$ $X_0 = B_t/\\|B_t\\|_F$ $X_{k+1} = \frac{3}{2}X_k - \frac{1}{2}X_kX_k^TX_k$ $W_{t+1} = W_t - \eta \cdot X_K$	• Momentum $\mu \in [0,1)$ • K iterations (typically 3-5) • Singular values in convergence basin	• Approximates UV^T cheaply • Momentum + geometry • Scalable	$O(Kd^2)$ per step ($K \ll d$)
Newton-Schulz (Polynomial)	Iterative convergence: $\Sigma_k \to I$ as $k \to \infty$	**Classic polynomial:** $p(\sigma) = \frac{3}{2}\sigma - \frac{1}{2}\sigma^3$ **Practical (NanoGPT):** $f(\sigma) = 3.4442\sigma - 4.7750\sigma^3 + 2.0315\sigma^5$	**Matrix iteration:** $X_{k+1} = \frac{3}{2}X_k - \frac{1}{2}X_kX_k^TX_k$ **Result:** If $X_k = U\Sigma_kV^T$, then $X_{k+1} = U \cdot p(\Sigma_k) \cdot V^T$	• $\sigma \in (0,1)$ for convergence • Fixed point: $p(1) = 1$	• Preserves singular vectors • Only transforms singular values • Avoids explicit SVD	$O(Kd^2)$